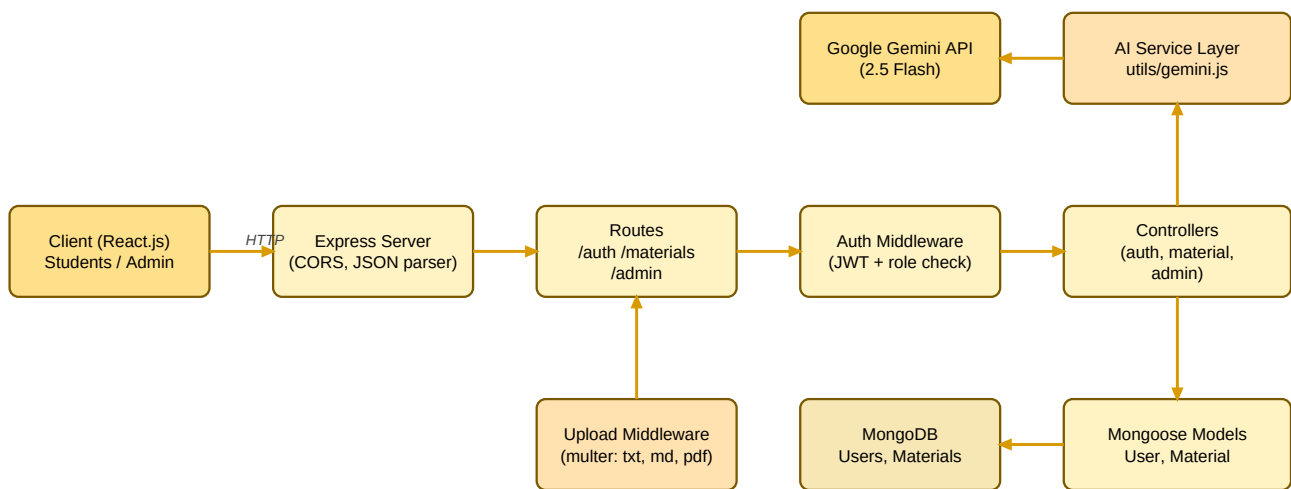


## Project Design Phase-II

# Technology Stack (Architecture & Stack)

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| Date          | 29/09/2026                                                          |
| Team ID       | SWTID-2026-7994                                                     |
| Project Name  | AI StudyBuddy - AI assistant for smarter learning and study support |
| Maximum Marks | -                                                                   |

## Technical Architecture:



Technical architecture of AI StudyBuddy

**Table-1: Components & Technologies**

| S.No | Component              | Description                                                        | Technology                                                    |
|------|------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|
| 1    | User Interface         | Web client where students and admins interact with the application | React.js (frontend); Postman / Thunder Client for API testing |
| 2    | Application Logic-1    | Server, routing, controllers and middleware                        | Node.js, Express.js                                           |
| 3    | Application Logic-2    | Authentication and authorization                                   | JWT (jsonwebtoken), bcryptjs, role check middleware           |
| 4    | Application Logic-3    | Study material upload handling                                     | multer (.txt, .md, .pdf, max 5 MB)                            |
| 5    | Database               | Users and study materials with embedded AI resources               | MongoDB with Mongoose ODM                                     |
| 6    | Cloud Database         | Managed database hosting                                           | MongoDB Atlas                                                 |
| 7    | File Storage           | Uploaded files saved to the uploads folder on the server           | Local filesystem (uploads/)                                   |
| 8    | External API           | Generates educational content from study material                  | Google Gemini API (2.5 Flash) via @google/genai               |
| 9    | Machine Learning Model | Large language model for summaries, flashcards, quizzes and plans  | Google Gemini 2.5 Flash                                       |

| S.No | Component                       | Description                                                                                                                            | Technology                                          |
|------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 10   | Infrastructure (Server / Cloud) | Local: Node.js v16+, npm v8+, Windows/macOS/Linux, 8 GB RAM (16 GB recommended), 1 GB disk. Cloud: any Node.js host with MongoDB Atlas | Local server (PORT 5000); cloud deployment optional |

**Table-2: Application Characteristics**

| S.No | Characteristics          | Description                                                                                              | Technology                                                   |
|------|--------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1    | Open-Source Frameworks   | Open-source runtime, framework, ODM and libraries                                                        | Node.js, Express.js, Mongoose, cors, dotenv, multer, nodemon |
| 2    | Security Implementations | Stateless auth, password hashing, role-based access, upload validation, secrets in environment variables | JWT, bcryptjs, RBAC, .env, CORS                              |
| 3    | Scalable Architecture    | Modular MVC layers with an isolated AI service                                                           | Routes / controllers / models / middleware / utils           |
| 4    | Availability             | Stateless server instances behind a load balancer with a managed database                                | Load balancer (deployment option), MongoDB Atlas             |
| 5    | Performance              | Asynchronous AI calls, upload size limit and lightweight routing                                         | Express, async/await, multer limits                          |